

1) Unsupervised learning is when our data is not explicitly labeled, and we ask the algorithm to learn something meaningful on its own without labels.

2) The 2 types of unsupervised learning we will explore today are: K-means and Principal component analysis

3) What is a centroid? Take some time on this question after reading under the K-Means block in the ipynb. Draw some of your understanding out on the board! Include some formulas if you can.

A centroid is the mean/average point

4) Under line 7, the blue points represent data points and the stars represent centroids.

5) Write the formula for finding the closest centroids on the board, and then summarize it using your own words here. Centroid: finds the distance between the position of the centroid and the point that is shortest

6) Read the code in line 8. Name the type of python object each of these are:

Centroids: float

X: 2d array

np.square: function

Min_distance: integer

Idx: float

7) What is the purpose of line 9?

Line 9 checks if the function works

8) What is the difference between the first plot under line 7 versus the one in line 10?

First plot under line 7 is the plot of initial centroids and the datapoints

The one in line 10 is the same thing but color coded - meaning each datapoint is assigned to a centroid

9) On the board, write the equation under 'Computing Centroid Means' and label each of the variables. On your canvas submission, write 2 sentences on what this equation represents.

This equation represents how the kmeans unsupervised learning method assesses the importance/weight of a point and how it influences predictions. It uses the datapoints and calculates the distance from the centroid (by seeing which cluster it's in) to weigh importance.

10) Challenge: What is the purpose of line 17-24?

Runs 10 iterations of your data to see how the closest centroids shift and how the cluster assignment correspondingly shifts

11) Line 25 onwards shows you how to do exactly what the above lines did with hard code. That means that scikit learn has an inbuilt function for all the things we just hand-coded! Yay! Write the code that helps you do what we did faster here:

```
from sklearn.cluster import KMeans
```

```
km1= KMeans(3)
```

```
km1.fit (X1)
```

12) Look up the parameters for Kmeans and explain them here!

n_clusters=how many clusters to create and how many centroids to create too - integer (Default is 8)

init=initialization method (a callable function, an array etc.) →k-means++ (default), random(chooses clusters and centroids at random), array (n_clusters,n_features), callable (X, n_clusters, random state= __)

13) Check out the image compression example from lines 27-36, and write at least 3 questions on the board that you have about it (if you don't have any questions, write 3 'takeaways'.

14) It's PCA time. What type of object is X3 in line 39? 2d array because it has two dimensions when you print the shape

15) What type of normalization are they doing for this data? Standard normalization

16) Vectors are kinda hard. Find a resource online to help you understand #3-- I'm here to help! Insert the resource here.

Geeks for geeks: Singular Value Decomposition (SVD) is a factorization method in linear algebra that decomposes a matrix into three other matrices, providing a way to represent data in terms of its singular values

17) In line 42, what is the purpose of this function, and what type of objects is the function returning?

Function is performing PCA → normalizes the data features, finds creates a variance matrix that measures how correlated those features are, and then computes eigenvectors

The function is returning numpy arrays.

18) Don't worry about all the particularities of line 47, data visualization is it's own class and specialty, but check out the graph. Read the paragraph just below and explain it in a sentence in normal-people speak here. I would ignore the paragraph after that unless you're feeling brave.

The black and green lines represent where the most spread and accounts for where the most variation lies

19) Skip ahead to the graph under line 54. Write a sentence about what this is doing for me.

The graph under line 54 shows how projecting the data to a lower dimension moves the position of the data

20) Lines 55 do the thing that happened in the above lines for PCA with an in-built function in scikit learn. Insert the code here:

```
from sklearn.decomposition import PCA
```

```
pca= PCA (n_components= 2)
```

```
pca_result= pca.fit_transform(X3)
```

21) In the code you just wrote above, where did X3 even come from? What is it? And what is n_components anyway?

X3 came from line 39 and is essentially the X column (or the column called X) of the dataset they imported. N_components means how many principal components you want to keep after PCA / shrinking your features down.

22) Somewhere before the graph are a few sentences that I would like you to comprehend:

```
pca_result: (50, 2)  
Variation per principal component: [0.87062385 0.12937615]
```

The above ratio is how much of the variance or variability in our data is captured by the two principal components.

You may have to look this up, but what does the (50, 2) mean? And in normal people speak, what does the output for Variation per principal component mean?

It means how much of variation of the data each PCA component accounts for (together the PCA components add up to 100%).

50,2 is the shape of the numpy array that is a result of performing PCA on the data (2 dimensional).

23) Skip ahead to line 89, and read through the code, it's kinda cool!

I read it

24) Here's the fun now... Import the important libraries you need in a fresh ipynb, and run this code:

```
from sklearn.datasets import load_sample_image
import matplotlib.pyplot as plt
```

```
china = load_sample_image("china.jpg")
plt.imshow(china)
plt.axis('off')
```

Can you try to compress this image for me, using K-Means?

25) Extra super bonus if you are already done: Find a way to upload your own (class appropriate) image into an ipynb, and see if you can reduce the number of colors in it.